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Sent via email to [REDACTED] on March 15, 2025.

March 15, 2025

Faisal D'Souza
Office of Science and Technology Policy
NCO, 2415 Eisenhower Avenue,
Alexandria, VA 22314, USA

Re: Comments of the National Center for the Advancement of Semiconductor Technology (Natcast) to Request for Information "Development of an Artificial Intelligence (AI) Action Plan," 90 Fed. Reg. 9088 (February 6, 2025)

Ref: Docket No. 2025-02305 (90 FR 8741)

Dear Mr. D'Souza,

On behalf of the National Center for the Advancement of Semiconductor Technology (Natcast), we are pleased to submit our response to the federal solicitation for the AI Action Plan.

Natcast is a purpose-built, non-profit entity designated to operate the National Semiconductor Technology Center (NSTC). Established by the bipartisan CHIPS and Science Act of the U.S. government, the NSTC is a public-private consortium dedicated to semiconductor R&D in the United States. The NSTC reflects a once-in-a-generation opportunity for the U.S. to drive the pace of innovation, set standards, and secure global leadership in semiconductor design and manufacturing.

The mission of the NSTC is to serve as the focal point for research and engineering throughout the semiconductor ecosystem, advancing and enabling disruptive innovation to provide U.S. leadership in the industries of the future.



We at Natcast recognize the critical importance of advancing artificial intelligence as a cornerstone of US technology leadership, national security, and economic competitiveness. Central to this mission is the work being done at the National Semiconductor Technology Center, which is pivotal to the United States' position in the global AI landscape and semiconductor technology.

The rapid growth of AI technologies relies heavily on robust semiconductor innovations- whichever nation leads in semiconductors is best positioned to lead in AI. As the President stated on March 3, 2025, "Semiconductors are the backbone of the 21st century economy...powering everything from AI to automobiles to advanced manufacturing."¹ Therefore, the NSTC plays an essential role in fostering transformative, high-risk/high-reward semiconductor R&D, on-shore in the United States, and will be critical in ushering in a "Second Bold Era in S&T,"² in this case, in artificial intelligence and semiconductors.

Over the past decades, as the semiconductor industry moved many operations out of the United States, much of this research and development moved offshore to other countries. To combat this, the NSTC is playing an essential role in onshoring semiconductor R&D, fostering new collaborative public-private projects that are driving needed advancements in semiconductor technology, which, in turn, fuels the development of cutting-edge AI technologies and applications here in the United States.

Further, by investing in semiconductor research and development, in partnership with America's leading experts, private industry, and academic centers of excellence, we are not only addressing current technological and infrastructure challenges but also paving the way for future breakthroughs that could lead to US pre-eminence in AI.

Finally, the NSTC provides a tremendous return on investment for the taxpayer, who will benefit from US AI leadership. For every dollar the US government spends on semiconductor R&D, that dollar results in *at least* \$15 in economic growth³ in the United States. Therefore, the NSTC promises to secure significant economic growth in the United States for the AI industries of the future.

¹ [Remarks by President Trump on Investment Announcement – The White House](#)

² [M-19-25, FY 2021 Administration Research and Development Budget Priorities](#)

³ [Sparkling Innovation - Semiconductor Industry Association](#)



Our submission includes the [National Semiconductor Technology Center Strategic Plan](#)⁴, outlining the NSTC vision and objectives to enhance semiconductor. The plan outlines a comprehensive strategy for building the NSTC as an enduring institution that will strengthen and extend U.S. leadership in semiconductor technology. It outlines the NSTC's three primary goals:

1. Extend U.S. technology leadership
2. Reduce the time and cost to prototype
3. Build and sustain a semiconductor workforce ecosystem

Based on the NSTC Strategic Plan, we share the following specific efforts, addressing specific challenges essential to the AI Action Plan:

1. **Hardware Innovations have Historically High Development Costs, High Research Infrastructure Costs, Expensive and Specialized Expertise, and Long Timelines:** Companies, startups, and venture capital firms are often hesitant to invest in semiconductor technologies that are capital-intensive and have lengthy development cycles. Consequently, many promising semiconductor start-ups, including those focused on manufacturing process innovation such as wafer foundry, metrology, and equipment, face difficulties attracting private investment.
 - o **NSTC Strategic Initiatives:**
 - The NSTC is moving quickly to establish a best-in-globe set of research infrastructure assets that lower the barriers to semiconductor prototyping, experimentation, and other R&D activities. The facilities will allow innovators to collaborate and accelerate solving the most challenging problems in semiconductors related to AI and next generation computing. The facilities will offer unparalleled value to stakeholders across the semiconductor and AI value chain. The facilities announced to date are the following:
 1. NSTC Prototyping and NAPMP Advanced Packaging Piloting Facility in Phoenix, Arizona.
 2. NSTC Administrative and Design Facility (ADF) in Sunnyvale, California.
 3. NSTC Extreme Ultraviolet (EUV) Accelerator in Albany, New York.

⁴ [NSTC Strategic Plan | Natcast](#)



- Natcast has created an NSTC investment fund to support promising semiconductor ecosystem start-ups in the United States, including in the AI and datacenter space.
 1. The NSTC investment fund will fill this market gap by strategically investing in start-ups that align with the research agenda or have true breakthrough potential. The fund's objective is to provide capital and an ecosystem that enables start-ups to achieve milestones and positions them to attract private capital.
 2. Natcast will also provide start-ups with access to prototyping and advanced packaging facilities, design tools, and silicon aggregation services.
- 2. **Hardware is a Key Bottleneck and Enabler of AI Competitiveness and Energy Efficiency:**

Absolute performance is essential for frontier model leadership, while energy efficiency is crucial for commercial adoption. The U.S. supply chain must continue to provide processors, communications, and memory solutions with leading-edge technology. This requires best-in-the-world, new, R&D paradigms and research infrastructure, which the NSTC is poised to supply here in the United States.

 - o **NSTC Strategic Initiatives:**
 - Supporting this infrastructure investment, the NSTC's R&D Technology Advisory Board and research agenda will have the world's leading semiconductor experts and AI innovators providing input on focus areas that promise to result in transformative, high-risk/high-reward R&D. These projects have the potential to generate breakthrough innovations onshore, in the United States.
 1. NSTC's research agenda will ensure that U.S.-originated AI datacenter and edge system technologies lead in absolute performance (e.g., with beyond-Complementary Field-Effect Transistor (CFET) logic devices, integrated silicon photonics and chip-to-chip communications, new high-performance universal storage).
 2. NSTC will also support computing performance per watt through system-level optimizations across logic, connectivity, and memory.

3. **Resilience of Domestic Semiconductor Supply Chains:** Domestic semiconductor supply chains, and the resulting AI supply chain, need to be immune from nation-state disruption, and the U.S. needs to be able to deploy U.S.-based semiconductor and AI innovations.

- o **NSTC Strategic Initiatives:**

- NSTC's research into cost-effective manufacturing techniques and waste treatment technologies will address the economics of fab/factory construction and enable the economically viable onshoring of the currently global semiconductor supply chain, avoiding potential costly disruptions.
- NSTC's research administration policies- emphasizing research security, domestic production, intellectual property - combine to enable onshore capacity for leading technologies critical to AI, while simultaneously building in security and U.S. sourcing from the ground-up.

4. **Whole-of-Sector Response for Effective AI Leadership:** Effective AI leadership requires a whole-of-sector response across the U.S. semiconductor industry.

- o **NSTC Strategic Initiatives:**

- NSTC's growing membership, currently 113+ members⁵, comprises the full spectrum of participants and stakeholders in the semiconductor industry, from academic institutions to tool vendors to design companies to manufacturers and even end users.
- By convening a whole-of-sector R&D capability, the NSTC will facilitate faster adoption of key semiconductor and AI technologies, and foster coordination on approaches and standards in areas ranging from chiplets to supply chain integrity.

⁵ [NSTC Members | Natcast](#)



5. **Workforce Preparation:** To be competitive, the U.S. needs to prepare the workforce for AI-related jobs in the semiconductor industry.

o **NSTC Strategic Initiatives:**

- Through the NSTC Workforce Center of Excellence, the NSTC is developing and funding training and education programs for the semiconductor industry and fostering partnerships between educational institutions and AI companies to ensure relevant and up-to-date training.
- Over the next ten years, the WCoE aims to enable over 35,000 individuals for roles in the semiconductor industry. During its first year of operation, the WCoE expects to:
 1. Enable training of 2,000 individuals for roles in the industry
 2. Serve more than 30 stakeholder institutions, including all leading-edge manufacturing employers

Thank you for the opportunity to comment.

We look forward to the opportunity to contribute to the AI Action Plan and further the important work of the NSTC in driving US AI leadership.

Sincerely,

Maryam Cope
Vice President, Governmental Affairs
Natcast, Operator of the NSTC